KisanSahay

CLINE AI BUILDER HACKATHON

Track: Agriculture | IIT Guwahati

PROBLEM STATEMENT

PROJECT OVERVIEW

Project: **KisanSahay-AI**

A voice-first help for seven farmer problems, one shareable **Action Plan**

Track / Category: Agriculture | Cline AI Builder Hackathon

Team Name: hello

The Challenge:

Schemes | Claims | Soil | Mandi / Market | Livestock | Crop | Weather



KisanSahay-AI Team

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IIT Guwahati

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PROBLEM

CORE IDEA

Shared FarmerProfile

Reuses farmer context across modules; fields can be corrected before action.

Seven module runner

Same result contract, with Deep / Working / Prototype labels.

Intent router

Routes the farmer's issue to one or more relevant modules.

Cline runtime

Rule ingestion, generated tests, form-fill and validation behind an adapter.

Shareable output

PDF, WhatsApp-ready text and JSON; cached plan for offline reopen.

Action Plan builder

Combines priorities, deadlines, documents and cross-module notes.

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PROBLEM

PROBLEMS ADDRESSED

Shared farmer profile

One interview captures core context once and passes it to relevant modules.

Traceable outputs

Each module returns actions, sources, provenance and missing fields in a common contract.

Safety constraints

Eligibility and deadlines use deterministic rules; triage escalates red flags to experts.

Seven-module reach

Deep: Schemes, Claim. **Working:** Soil, Mandi. **Prototype:** Crop, Advisory, Livestock.

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SOLUTION

INTAKE & PROFILE

Voice, typing or tap chips

A short intake builds a shared FarmerProfile: language, location, land, crop/season and module-specific details. Optional photo and GPS fields are confirmable.

ROUTE REQUEST

Intent routing & confirmation

The router selects one or more modules and prioritizes time-critical tasks. Farmers can correct key values before actions.

MODULE RUNNERS

Run modules to one contract

Each module returns a Zod-validated ModuleResult: status, structured body, actions, sources, provenance, safeguards and missing fields.

BUILD ACTION PLAN

Build the Action Plan

The deterministic builder orders actions by priority and due time, merges document checklists and cross-module notes, and remains the fallback when orchestration is unavailable.

SHARE / REOPEN

Dossier & sharing

Render PDF, WhatsApp-ready text and JSON. The latest plan can be reopened offline; KisanSahay-AI does not submit forms or claims on the farmer's behalf.

ACTION PLAN OUTPUT

Usable result, not just chat

A single checklist shows what to do now, what to collect, what to wait for and where to seek expert help.

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SOLUTION

CLINE RUNTIME WORKFLOWS

Ingest a circular

- Extract a cited draft rule
- **Generate tests from examples**
- Run tests + repair (max 3 loops)
- Keep draft until human approval

Adapter + validators

- **Call SDK via ClineAdapter**
- Schema-constrained output
- Plain-code validation
- Fixture fallback for replay

Runtime tasks

- Rule updates + reviewable diffs
- Form-fill + claim validation
- **Optional plan orchestration**
- **Deterministic builder remains fallback**

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INNOVATION

CORE LAYERS

LANGUAGE & PROFILE

RULES-AS-DATA

CLINE RUNTIME

OUTPUT & SAFETY

INNOVATION & DIFFERENTIATORS

One profile, seven modules

A shared FarmerProfile feeds the seven pathways; each returns a consistent, source-carrying ModuleResult.

Rules-as-data + generated tests

Cline drafts a rule, generates example-driven tests, runs validation and proposes repairs. Human approval is required before publication.

ModuleResult + Zod

One typed envelope carries status, actions, source links, provenance, required fields and a safety disclaimer.

Constrained triage

The LLM extracts fixed-vocabulary features; deterministic scoring and red-flag overrides return candidates and urgency, not diagnosis.

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INNOVATION

TECHNICAL FEASIBILITY

Source-backed rules

Scheme records carry source URLs and verification dates; current rules and deadlines must be checked before display.

Deterministic logic

Code/data evaluate eligibility, stacking, deadlines, soil bands and mandi net returns; LLM explanations do not determine verdicts.

Cline adapter

ClineAdapter isolates the runtime SDK; fixture fallback supports predictable replay when needed.

Prototype stack & tiers

React/Vite/TypeScript/Tailwind PWA + Node/Fastify/Zod. Schemes/Claim are Deep; Soil/Mandi Working; Crop/Advisory/Livestock Prototype.

IMPLEMENTATION PATHWAY



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MODEL ARCHITECTURE

Core Engines & Runtime

INTENT LAYER

LLM extracts user intent and fixed-vocabulary crop/livestock features into low-temperature structured JSON.

Deterministic Core:

TypeScript evaluates scheme rules, stacking, claim deadlines, soil bands, mandi net returns and triage red flags. The LLM does not decide eligibility.

Shared Profile & Contracts

SHARED FARMER PROFILE

Language, district, land, crops, season, category, known benefits, loss event and task-specific inputs. Photo fields and GPS are optional and confirmable.

UNIFORM MODULE RESULT

Every module returns status, structured body, actions, sources, provenance, safeguards and missing fields through one Zod-validated contract.

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MODEL ARCHITECTURE

Deterministic Engines

DETERMINISTIC RULES

Code/data evaluate eligibility, scheme stacking, deadlines, soil bands and mandi net-of-transport comparisons; the LLM explains but does not decide.

SAFE TRIAGE

Crop and livestock modules return likely candidates, safe steps and urgency. Fixed vocabularies and red-flag overrides escalate risk; no pesticide or drug doses.

Cline Runtime Workflows

Cline Runtime:

ClineAdapter wraps runtime tasks for rule ingestion, generated tests, form-fill and validation. Structured output is checked by plain-code validators.

Human review:

New rules stay in draft until a reviewer approves them; the optional orchestration agent falls back to the deterministic Action Plan builder.

The challenge is keeping seven pathways consistent and safe. KisanSahay-AI combines one farmer profile, typed module results and source-linked next steps—with deterministic decisions and human review.

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IMPACT AND BENEFITS

PESTEL ANALYSIS

P POLITICAL

Scheme pathways need current official rules and local-state verification before publication.

E ECONOMIC

Potentially reduces time spent searching portals and assembling documents; savings are not measured.

S SOCIAL

Assamese voice tested; Hindi/English and typed/chip fallbacks are in the product design. Field performance is not quantified.

T TECHNOLOGICAL

Seven modules share one profile and typed contract; deterministic rules bound high-stakes decisions.

E ENVIRONMENTAL

Timely weather/crop guidance may help avoid losses; no measured crop, soil or water outcomes reported.

L LEGAL

Request consent for photo/GPS capture, confirm sensitive fields and prepare drafts only; no automatic portal submission.

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IMPACT AND BENEFITS

POLICY & SCALABILITY ALIGNMENT

Official scheme data

Scheme Finder organizes PM-KISAN, PMFBY and KCC; show only sourced, dated rules.

Regional access

Assamese voice tested; local WER, latency and noisy-field performance remain unmeasured.

Service delivery

PDF, print, WhatsApp-ready text and offline reopen support CSC / extension workflows.

Maintenance at scale

Rules-as-data, sources, generated tests and human review; local coverage needs validation.

SCALE CHECKS

PUBLIC SCHEME DATA

Scheme Finder organizes PM-KISAN, PMFBY, KCC and more; every current rule needs a dated official source.

LANGUAGE ACCESS

Assamese voice tested; local accuracy, latency and noisy-field robustness are not measured.

SERVICE DELIVERY

PDF, print, WhatsApp text and offline reopen can support CSC and extension workflows.

MAINTAINABLE SCALE

Rules-as-data, source trails, generated tests and human approval; state/district coverage needs validation.

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IMPACT AND BENEFITS

IMPACT ON STAKEHOLDERS

FARMERS

Plain-language steps, document checklist and deadline prompts may reduce friction.

CSC OPERATORS

One dossier can help staff organize intake and explain the next action.

EXTENSION WORKERS

Sourced referrals for crop, weather and livestock questions; not expert replacement.

RESEARCH & REVIEW

Reviewable rules and tests support upkeep; outcome evidence still has to be collected.

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REFERENCES & ACTION PLAN VIEW

REFERENCES

[1] PROJECT SOURCE — KisanSahay-AI Blueprint v2 + Technical Architecture & Execution Guide v2. These define intended architecture and tiering, not independent outcome validation.

[2] OFFICIAL SCHEME SOURCES — PM-KISAN: pmkisan.gov.in • PMFBY: pmfby.gov.in • Soil Health Card: soilhealth.dac.gov.in. Recheck current rules, dates and local channels.

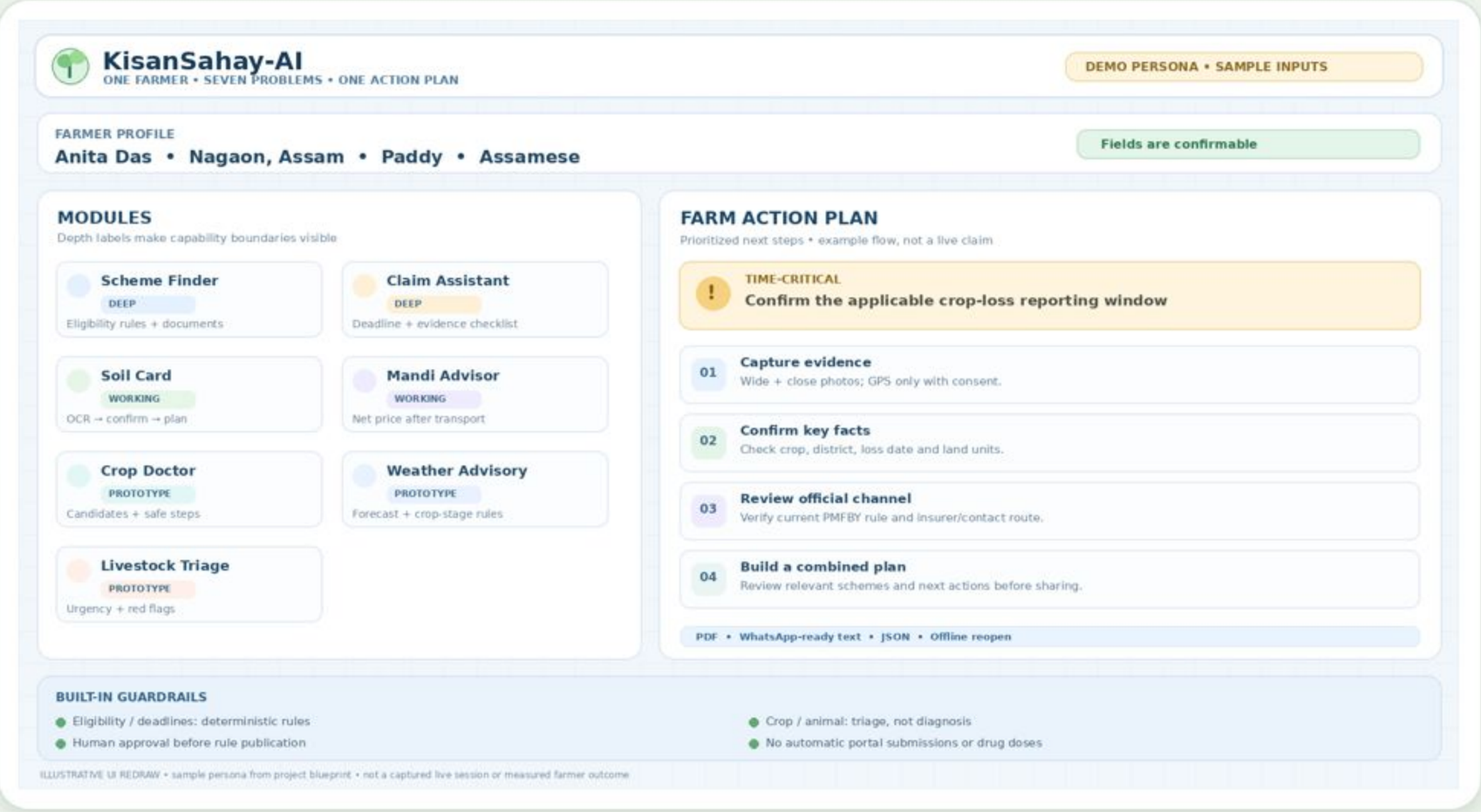
[3] DATA & WEATHER — data.gov.in (market-price datasets) • Open-Meteo API docs: open-meteo.com/en/docs. Report data date, missing coverage and cached/stale status.

[4] RESEARCH CAVEAT — Mohanty et al. (2016), "Using Deep Learning for Image-Based Plant Disease Detection," Frontiers in Plant Science, DOI: 10.3389/fpls.2016.01419. Historical benchmark results do not establish KisanSahay accuracy.

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REFERENCES & ACTION PLAN VIEW

ACTION PLAN



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THANK YOU

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KisanSahay-AI | One farmer, seven problems, one Action Plan